

Decision theory unbound

Zachary Goodsell

Department of Philosophy, University of Southern California, Los Angeles

Correspondence

Zachary Goodsell, Department of Philosophy, University of Southern California, 3709 Trousdale Pkwy, Los Angeles, CA, 90089.
Email: zgoodsel@usc.edu

Abstract

Countenancing unbounded utility in ethics gives rise to deep puzzles in formal decision theory. Here, these puzzles are taken as an invitation to assess the most fundamental principles relating probability and value, with the aim of demonstrating that unbounded utility in ethics is compatible with a desirable decision theory. The resulting theory frames further discussion of Expected Utility Theory and of principles concerning symmetries of utility.

1 | THE PARADOX

The *utility* of an outcome is a number which measures how much a chance of that outcome contributes to the overall value of a risky gamble. For example, an outcome has utility 2 just in case a fifty-fifty gamble between that outcome and one with utility 0 is equally good as certainty of an outcome with utility 1. More generally, an outcome has utility x , for x a number greater than 1, if and only if a gamble with a $1/x$ chance of that outcome, that otherwise yields utility 0, is equally good as a sure utility 1. If for every number some possible outcome has a utility at least as great as that number, we say that utility is *unbounded*.

If you read “good” above as “morally good”, then whether utility is unbounded in that sense is a substantial ethical question. For example, one might think that it is good to create people whose lives will be wonderful, and so one might wonder whether a $1/n$ chance of creating n such people is equally good as creating exactly one such person for sure. For this to be so requires utility to be unbounded. If you read “good” in some other way, such as “good for me” or “good for the sake of winning this chess game”, then whether utility is unbounded in the concomitant sense is a different question, but an equally substantial one in each case. Most of what follows will be true

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2023 The Authors. *Noûs* published by Wiley Periodicals LLC.

TABLE 1 The first-toss St Petersburg prospect.

T	HT	...	H...HT	...
2	4	...	2 ⁿ	...

on many reasonable disambiguations, but where ambiguity might be problematic we will default to the moral reading.

The problem motivating this paper is that there are significant structural problems for the thesis of unbounded utility, which threaten to refute the thesis before the substantial ethics can even begin. One widely recognised problem is that, when utility is unbounded, the *expected utility* of a prospect might be undefined. In that case, standard decision theory, which compares prospects by comparing their expected utilities, falls silent. The canonical witness to this fact is the *St Petersburg prospect*, which yields utility 2 with probability 1/2, utility 4 with probability 1/4, and in general utility 2ⁿ with probability 1/2ⁿ. The expected utility would be given by the sum

$$\frac{1}{2} \times 2 + \frac{1}{4} \times 4 + \dots + \frac{1}{2^n} \times 2^n + \dots = 1 + 1 + \dots + 1 + \dots$$

Since the sum does not converge, the expected utility is undefined. As a result, standard decision theory does not say how to value the St Petersburg prospect. In particular, it does not yield the correct verdict that it is worse than a prospect that with probability 1/2ⁿ yields utility 2ⁿ + 1 instead of 2ⁿ. This defect remains even when the expectation is taken in the extended real numbers, and hence can be given the value ∞. In that case, both of the prospects get expected utility ∞, so expected utility still cannot distinguish between them.¹

But the problem is significantly worse than a mere gap in standard decision theory. The hypothesis of unbounded utility is flatly inconsistent with some very compelling principles governing the comparison of prospects, whether or not expected utilities are assigned to them. The question is not how to extend standard decision theory in this or that way, but rather whether a reasonable decision theory is compatible with unbounded utility at all. If not, then ethical principles that imply unbounded utility crash immediately.

The problem—a version of the Two-Envelope paradox—will be explained informally in the remainder of this section, and is precisified in Section 2. Section 3.3 presents and motivates a theory for reasoning in the presence of unbounded utility and establishes its consistency, thus guaranteeing that no further paradox arises. Having established the viability of this theory, the paper concludes (Section 4) by pointing out two of many interesting and underexplored avenues for further research on decision theory in the presence of unbounded utility.

The motivating paradox of the paper, first presented in Nalebuff (1989), can be illustrated as follows.² Suppose a coin is to be tossed (fairly and independently) infinitely many times. You have two options. The first is to take the *first-toss St Petersburg prospect*, whose outcome is determined by how the coin lands in accordance with Table 1. Choosing this prospect yields utility 2 if the coin lands tails right away, utility 4 if it lands heads then tails, 8 if it lands heads-heads-tails, and in general 2ⁿ if it takes *n* tosses for it to land tails for the first time. Your second option is to take the *second-toss St Petersburg prospect*, described in Table 2. This prospect differs only in

¹ See Nover and Hájek (2004) for further discussion.

² See also Broome (1995), Arntzenius and McCarthy (1997), McGee (1999), Chalmers (2002), and Dietrich and List (2005). A similar puzzle is found in Seidenfeld et al. (2009).

TABLE 2 The second-toss St Petersburg prospect (“/” separates the first coin toss from the rest).

T/T	H/T	T/HT	H/HT	...	T/H...HT	H/H...HT	...
2	2	4	4	...	2^n	2^n	...

TABLE 3 The first-toss prospect redescribed.

T/T	H/T	T/HT	H/HT	...	T/H...HT	H/H...HT	...
2	4	2	8	...	2	2^{n+1}	...

that the first coin toss is ignored. So whether the first toss lands heads or tails, the second-toss prospect yields utility 2 if the second toss lands tails, utility 4 if the second lands heads and the third lands tails, and in general utility 2^n if, starting from the second toss, it takes n tosses to get tails for the first time.³ Which prospect, if either, is better? Here are arguments to two contradictory answers:

First argument: the two prospects are *equally good*. When evaluating a risky prospect, which coin tosses determine the outcome is not relevant. What matters are the probabilities of the various outcomes and the utility of those outcomes, and the two prospects are equal in these respects.

Second argument: the first-toss prospect is *strictly better* than the second-toss prospect. For suppose you learn the following: after the first toss, it took another n tosses for a toss to come up tails. That is, you learn something of the form: either T/H...HT or H/H...HT. This reduces your decision to that between the green cell in Table 3 and the green cell in Table 2. And the green cell in Table 3 is better: halfway between 2 and 2^{n+1} is $2^n + 1$, so fifty-fifty between 2 and 2^{n+1} is an improvement on a sure 2^n . But this is true for any n , so we can know in advance that the first will look a better gamble than the second. It seems then we should cut to the chase and judge the first better unconditionally.

These contradictory arguments constitute the paradox. The easy way out is to deny the possibility of the relevant prospects, an option usually taken by denying the unboundedness of utility.⁴ Here, we will be taking the hard way out, which is to accept the possibility of the prospects but to block the reasoning that led to a contradiction.⁵ We will find that it is the second argument that must go. Although for any n the first-toss prospect is better than the second-toss prospect conditional on it taking n tosses to get tails, it is a fallacy to conclude from this that the first-toss prospect is better unconditionally (or so the proponent of unbounded utility must say). But this is only the beginning of the solution. The challenge is in demonstrating that, even without this intuitively tempting principle, there is a strong and systematic decision theory to be had that makes room for unbounded utility.

³ If it matters, we can stipulate that both prospects yield utility 2 in the probability zero event that all tosses land heads.

⁴ Savage (1954): 80–82, Russell and Isaacs (2021).

⁵ Broome (1995) suggests this line, and it is endorsed by Arntzenius and McCarthy (1997), Dietrich and List (2005), and Meacham (2019), among others.

2 | FORMALIZING THE PARADOX

2.1 | Primitives

An *outcome* is intuitively a situation you might end up in, for which there is no further uncertainty about how good things will turn out if that outcome obtains. A *prospect* (also called a *gamble* or an *act*) is intuitively an option available to you which yields certain outcomes with certain probabilities. The object of decision theory is to rank the prospects in terms of the values of the outcomes they might yield and the probabilities with which they yield them.

To get clear on the paradox, it will be useful to employ a standard formalization of decision theory due to Savage (1954), which is rich enough to bring out the paradox, but idealized enough to obscure irrelevant complexities.

For concreteness, we can think of prospects as different ways of gambling on the result of the infinite sequence of coin tosses introduced in Section 1. That is, a prospect associates an outcome with each way the coin tosses might land, specifically the outcome that you will end up in if you take the prospect and the coin tosses land that way. Thus, the probability of a prospect p yielding an outcome o is given by the probability that the coin tosses land in one of the ways that p associates with o .

Let Ω be the set of maximally specific ways the coin tosses might land,⁶ and $\mathbf{P}$ the usual probability function for fair and independent coin tosses (e.g., it gives a $1/4$ probability to the claim that the first toss lands heads and the third lands tails, and so on).⁷ We take as primitive the following:

- A space of *outcomes* $\mathcal{O}$ (note: outcomes should not be conflated with ways the coin tosses might land, since outcomes are supposed to be maximally specific with regards to how good things turn out).
- A space of *prospects*, $\mathcal{P}$, which is a set of functions from Ω to $\mathcal{O}$.⁸
- A relation $\succsim$ on prospects called *at least as good as*.

We also use $\succsim$ to compare outcomes with each other and with prospects, by identifying each outcome o with the prospect that yields o no matter what (i.e., the function that maps everything in Ω to o).

Finally, we define utility in the standard way, relative to an arbitrary choice of outcomes o_0 and o_1 , with $o_0 < o_1$. o_0 and o_1 respectively have utilities 0 and 1 by stipulation; then define the utility

⁶ Ω consists of propositions of the form

Toss 1 lands heads and ... and toss n lands tails and ...

These are standardly represented by the space 2^ω of infinite 0-1-valued sequences. Any other state space with sufficiently rich probability structure would also serve for present purposes.

⁷ With the usual measurable structure on 2^ω . $\mathbf{P}$ is the completion of the measure which gives probability $1/2^n$ to each set of sequences that contains all and only those which agree on some n positions.

⁸ We assume that $\mathcal{P}$ contains only measurable functions, where $\mathcal{O}$ is given measurable structure compatible with $\succsim$ restricted to constant functions. We say that a prospect p yields o in the state $x \in \Omega$ if p maps x to o . Moreover, the probability with which p yields an outcome in X is given by

$$\mathbf{P}(\{\omega \in \Omega : p(\omega) \in X\}) \quad (1)$$

function u so that a prospect with a $1/3$ chance of utility 3 and a $2/3$ chance of utility 0 is equally good as a sure utility 1, and similarly for other numbers.⁹

2.2 | Assumptions

The paradox can be illustrated by deriving a contradiction from seven widely accepted decision-theoretic assumptions. Four of these are axioms from Savage's *Foundations of Statistics* Savage (1954). The first two of these concern the very general structure of $\succsim$:

Preordering *At least as good as* is reflexive ($p \succsim p$) and transitive (if $p \succsim q \succsim r$, then $p \succsim r$).

Totality For any prospects p and q , either $p \succsim q$ or $q \succsim p$.

The next one specifies that every possible correlation between coin-toss sequences and outcomes in $\mathcal{O}$ corresponds to a prospect in $\mathcal{P}$:

Prospect Richness For any function¹⁰ f from Ω to $\mathcal{O}$, there is a prospect in $\mathcal{P}$ that yields $f(s)$ when the coin tosses land in the sequence s .

The key axiom of Savage is the *Sure-Thing Principle*, which relies on a defined notion of *conditional* comparison of prospects. For a given subset of coin toss sequences E , let $p \upharpoonright E$ be the prospect that yields the same outcome as p when the way the coin tosses land is in E , but which yields utility 0 in every other situation. Write $p \succsim_E q$ for the comparison $p \upharpoonright E \succsim q \upharpoonright E$. Savage's Sure-Thing Principle says that to compare prospects that are equally good conditional on E , you can ignore what happens on E and compare them conditional on the complement of E ($\Omega \setminus E$):

Sure-Thing Let E be a subset of Ω with probability strictly between zero and one such that p and q are equally good conditional on E (i.e., $p \sim_E q$). Then $p \succsim q$ if and only if $p \succsim_{\Omega \setminus E} q$.

The next three principles go beyond Savage's axioms. Each has also been endorsed at various points in the literature, but together they give rise to the paradox. The first is the assumption of unbounded utility among outcomes in $\mathcal{O}$.

Unbounded Utility For any real number x , there is an outcome in $\mathcal{O}$ that has utility x (§2.1).

⁹ In general:

- If $x \geq 1$, then for o to have utility x is for a $1/x$ chance of o and $1 - 1/x$ chance of o_0 to be equally good as a sure o_1 .
- If $x \leq 0 < 1$, then for o to have utility x is for an x chance of o_1 and $1 - x$ chance of o_0 to be equally good as a sure o .
- If $x < 0$, then for o to have utility x is for a $1/(1 - x)$ chance of o and a $1 - (1/(1 - x))$ chance of o_1 to be equally good as a sure o_0 .

¹⁰ Strictly, we should restrict to *measurable* functions, where Ω is given its usual probability space structure and $\mathcal{O}$ is given measurable space structure compatible with the ordering induced by $\succsim$ on $\mathcal{O}$ by constant functions from Ω to $\mathcal{O}$.

Unbounded Utility implies that there are outcomes of utilities 2, 4, 8, and so on. Prospect richness says that putting these outcomes into Tables 1 through 3 gives you a corresponding prospect.¹¹

The next assumption is that which undergirds the “first argument” of Section 1, which says that the coin a St Petersburg prospect starts from does not matter and so the two prospects are equally good:

Stochastic Equivalence If for every outcome o , p is equally likely as q is to yield an outcome at least as good as o , then p is equally good as q .¹²

In effect, Stochastic Equivalence says that the comparison of prospects supervenes on the values of outcomes and the probabilities that the prospect yields them with. It can easily be confirmed that the first-toss and second-toss prospects of Section 1 both yield an outcome of utility greater than 2 with probability 1/2, greater than 4 with probability 1/4, and so on. So by Stochastic Equivalence they are equally good. Although Stochastic Equivalence goes beyond Savage’s axioms, it is an entirely standard assumption for decision theory in settings where prospects can have infinitely many outcomes¹³ and in cases where expected utilities are undefined on account of unbounded utility.¹⁴

The “second argument” in the paradox said that the first-toss prospect is strictly better than the second-toss prospect. The argument relied on the idea of comparing the two prospects conditional on a each member of a certain partition of how the coins might land. The principle needed to vindicate this reasoning is a generalization of Savage’s Sure-Thing Principle (the name is from Russell and Isaacs (2021), in Dietrich and List (2005) a similar principle is called *Event-Wise Dominance*):

Countable Sure-Thing Let $\mathcal{E}$ be a partition of Ω into propositions of nonzero probability. Then if some prospect p is at least as good as q conditional on every member of $\mathcal{E}$, then p is at least as good as q . If in addition there is some member of $\mathcal{E}$ conditional on which p is strictly better than q (i.e., $p \succ_E q$), then p is strictly better than q .

Countable Sure-Thing implies that the first-toss St Petersburg prospect is strictly better than the second-toss prospect. We reason as follows. First, set $\mathcal{E}$ to be the following partition of Ω :

E_1 := the second coin toss lands tails

E_2 := the second coin toss lands heads and the third lands tails

⋮

¹¹ Prospect richness also guarantees that for each outcome o in $\mathcal{O}$ there is a prospect that yields o no matter how the coin tosses land. Thus we can compare outcomes in $\mathcal{O}$ with prospects and each other by using that sure prospect as a proxy (so, for example, $o \succeq p$ means that the sure o prospect is at least as good as p).

¹² In symbols:

$$\forall o(\mathbf{P}\{x \in \Omega : p(x) \succeq o\} = \mathbf{P}\{x \in \Omega : q(x) \succeq o\} \rightarrow p \sim q)$$

¹³ See e.g., Blackwell and Girshick (1979) [1954]: 104, Joyce (1999): 26.

¹⁴ Dietrich and List (2005), Easwaran (2014a), Meacham (2019).

E_n := the second through $(n + 1)^{\text{st}}$ coin tosses land heads and the $(n + 2)^{\text{nd}}$ lands tails

⋮

E_n is the claim that if you take the first-toss St Petersburg prospect you will end up in the green cell in Table 3, and if you take the second-toss St Petersburg prospect you will end up in the green cell in Table 2. Reasoning using Savage's principles we may derive that the green cell in the first-toss prospect is better for any n , so $\mathcal{E}$ is a partition of propositions of nonzero probability such that conditional on any, the first-toss prospect is better. Countable Sure-Thing licenses the conclusion that the first-toss prospect is therefore better unconditionally. Thus we have derived the paradox from the aforementioned seven principles: Savage's axioms of Preordering, Totality, Prospect Richness, Sure-Thing, plus the three additional assumptions of Unbounded Utility, Stochastic Equivalence, and Countable Sure-Thing. Something has to give.

2.3 | Weakening totality and stochastic equivalence

Two of the most contentious principles in the derivation are Totality and Stochastic Equivalence. A related contradiction can be derived even when these principles are significantly weakened. The argument is slightly more complicated, but is dialectically significant because it sidesteps two frequent motivations for rejecting Totality and Stochastic Equivalence.

Most saliently, many authors are skeptical that these principles apply to prospects with infinitely many possible outcomes. Seidenfeld, Schervish, and Kadane (2009) and Lauwers and Vallentyne (2016); Lauwers and Vallentyne (2017) reject Stochastic Equivalence in such cases, and in particular they reject the equal value of the first- and second-coin St Petersburg prospects. Easwaran (2014a) Lauwers (2016), and Meacham (2019) believe that such prospects give rise to failures of Totality. Others—including Hare (2010), Bales, Cohen, and Handfield (2014), Schoenfield (2014)—have raised issues for Stochastic Equivalence in cases where Totality fails among the possible outcomes. So to reinforce the argument we shall employ weakenings of the two principles that avoid both complaints.

Let a *simple* prospect be one that can only yield finitely many different outcomes. And say that outcomes o and u are *comparable* if either $o \succeq u$ or $u \succeq o$. We will restrict both Totality and Stochastic Equivalence to simple prospects whose possible outcomes are all comparable to avoid both sorts of worry:

Restricted Totality If p and q are simple, and every pair of outcomes either might yield is comparable, then either $p \succeq q$ or $q \succeq p$.¹⁵

Restricted Stochastic Equivalence If p and q are simple, and every pair of outcomes either might yield is comparable, then if p and q yield the same outcomes with the same probabilities, then they are equally good.

To derive the paradox, we compare the first-toss and second-toss St Petersburg prospect with a further St Petersburg prospect that is determined by a disjoint infinite set of coin tosses. (To avoid positing these new coin tosses, we could have split the coin tosses into two infinite collections at

¹⁵ Restricted Totality entails that every pair of outcomes in $\mathcal{O}$ is comparable—those who expect incomparability among outcomes may treat $\mathcal{O}$ as a special comparable subclass of all outcomes.

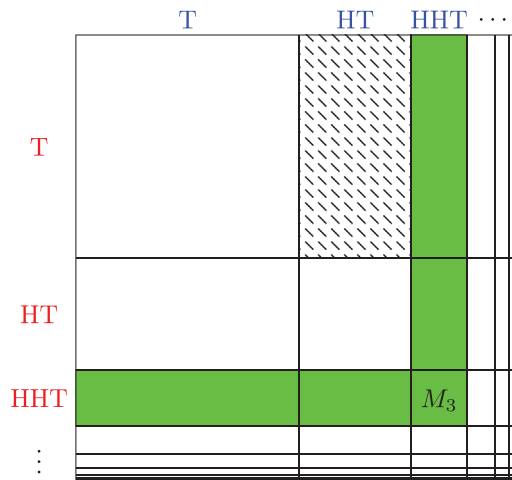


FIGURE 1 Geometric representation of state space with two infinite sequences of coin tosses. The space is partitioned into cells according to how many tosses it takes for the two sequences to get tails for the first time. For example, the hatched region represents the situation that the original sequence lands HT and the new sequence lands T. The proposition M_3 is highlighted in green. Note how the M_n will partition the space, and that each is symmetric with respect to the two sequences. [Color figure can be viewed at wileyonlinelibrary.com]

the beginning.) Call these the *new* coin tosses. The *new* St Petersburg prospect yields utility 2 if the first new coin toss lands tails, 4 if the first lands heads and the second tails, and so on.

Recall that, by countable Sure-Thing, the first-toss prospect is better than the second-toss prospect (the argument to this effect is not altered by the restriction of Totality and Stochastic Equivalence). The new paradox will be derived by showing that the new St Petersburg prospect is nevertheless equally good as both of those.

To this end, we partition the space of ways the tosses could land, Ω , as follows: let M_n be the proposition that either the original tosses or the new tosses took n tosses to land tails for the first time, and the other took no more than n (see Figure 1). This sequence of propositions forms a partition of Ω into propositions of nonzero probability, so to show the first-toss and new St Petersburg prospects are equally good we can show that they are equally good conditional on any member of this partition. Crucially, each of the first-toss and new St Petersburg Prospects is simple conditional on any M_n . The figure makes clear how the situation is probabilistically symmetric, so by Restricted Stochastic Equivalence the first-toss and new St Petersburg prospects are equally good conditional on any M_n . Countable Sure-Thing then yields their unconditional equal goodness. The same would have gone if we had replaced the original coin tosses from the beginning with the original coin tosses besides the first coin, thereby establishing that second-toss prospect is also equally good as the new prospect.

3 | UNBOUNDED UTILITY WITHOUT PARADOX

3.1 | Costs of rejecting countable sure-thing

Given that neither Stochastic Equivalence nor Totality is the culprit, it is most natural to view the paradox as demonstrating the incompatibility of Unbounded Utility with Countable

Sure-Thing. If so, then the viability of Unbounded Utility depends on whether Countable Sure-Thing can be rejected.

It must be stressed that Countable Sure-Thing formalizes an extremely tempting mode of reasoning in decision theory. This is the reasoning exemplified in Section 1, where we intuitively determine that the first-toss prospect is better than the second by dividing them each into cases (i.e., the green cell in Tables 2 and 3) and finding that first-toss prospect better conditional on any such case occurring. That is, Countable Sure-Thing gives voice to a sort of “anti-Dutch-book” principle: if you know you are about to learn something conditional on which the first-toss prospect looks better, then you should not need to learn it—you should be able to cut to the chase and judge it better unconditionally.

Failures of Countable Sure-Thing are counterintuitive, but it is unclear whether they should be regarded as impossible, or simply as one more surprising consequence of reasoning with the infinite (in this case, prospects with infinitely many possible outcomes). McGee (1999), Dietrich and List (2005), and Meacham (2019) urge that we should learn to live with the failure of Countable Sure-Thing, and Arntzenius, Elga, and Hawthorne (2004) reinforce the case by drawing parallels with the present paradox and a range of other puzzles in decision theory.¹⁶ It should also be recognized that Unbounded Utility also has much going in its favour, such as a wide range of general ethical principles that entail it.¹⁷

But there is another pressing problem for those who accept Unbounded Utility, which is that decision theory in the presence of Unbounded Utility is much less well understood than decision theory where Unbounded Utility is eschewed in favour of Countable Sure-Thing. In the latter case, all of the other principles listed in Section 2 can be consistently endorsed, and Russell (2020) has demonstrated that in the presence of these principles, Countable Sure-Thing gives rise to a highly general representation theorem (generalizing a result of Blackwell and Girshick (1979): 105). The result is an elegant, powerful, and overall desirable decision theory.

By contrast, work on decision theory in the presence of Unbounded Utility has proceeded in a very piecemeal fashion. Too often, the literature focuses on concrete methods of adding numbers together as proposals for avoiding the problem of undefined expected utilities, obscuring the more pressing need for deciding which of the fundamental principles of decision theory, such as Stochastic Equivalence or Savage’s Sure-Thing principle, can be accepted in light of the present paradox.¹⁸ When fundamental principles are discussed, the results often seem dire. Seidenfeld, Schervish, and Kadane (2009), as well as Lauwers and Vallentyne (2016); Lauwers and Vallentyne (2017) take the very drastic step of rejecting Stochastic Equivalence, whereas Fine (2008) and Smith (2014) question even the assumption that a prospect guaranteed to yield a better outcome is better. Another very common (although plausibly less problematic) concession is to accept widespread failures of Totality (see, e.g., Colyvan (2008), Easwaran (2014a), Lauwers (2016), Meacham (2019)).

Here is not the place to answer the question of which (if any) of Unbounded Utility and Countable Sure-Thing should ultimately be accepted. The goal of this paper is to address the second

¹⁶ See Russell and Isaacs (2021) for a response on behalf of Countable Sure-Thing.

¹⁷ Most notably, Unbounded Utility follows from standard Expected Total Utilitarianism, and therefore also from ethical views that agree with Expected Total Utilitarianism on a sufficiently wide range of case, e.g. in cases where deontological side-constraints are met or cases where all beneficiaries of an action are equally well-deserving. See also Beckstead and Thomas (2021), Russell (2021), Wilkinson (2022).

¹⁸ Especially literature following Nover and Hájek (2004), including Colyvan (2008), Easwaran (2008), Gwiazda (2014), Easwaran (2014b), and Bartha (2016).

problem, which is answered by demonstrating that even in the presence of Unbounded Utility, the highly orthodox principles listed above, besides Countable Sure-Thing but including Totality and Stochastic Equivalence, are consistent. Therefore, decision-theoretic orthodoxy need not be rejected to make room for Unbounded Utility.

3.2 | The theory DTU

This section compiles these principles with an important strengthening of Stochastic Equivalence to *Stochastic Dominance*, and illustrates how the resulting theory can capture a great deal of good decision-theoretic reasoning. The resulting theory will be called DTU, (for Decision Theory with Unbounded utility).

DTU extends the axioms stated in Section 2 with the exception of Countable Sure-Thing. These were the axioms of Savage, namely Preordering, Totality, Prospect Richness, and Sure-Thing, plus the two further axioms of Unbounded Utility and Stochastic Equivalence. DTU extends these axioms by strengthening Stochastic Equivalence to the also entirely standard principle of *Stochastic Dominance*:

Stochastic Dominance If for every outcome o , p is at least as likely as q is to yield an outcome better than o , then p is at least as good as q . If there is also some outcome o such that p is strictly more likely than q to yield an outcome better than o , then p is strictly better than q .¹⁹

Whereas Stochastic Equivalence says that the value of a prospect supervenes on the probability distribution over outcomes that prospect yields, stochastic dominance adds that shifting probability from worse outcomes to better ones is always an improvement. DTU will also include the simplifying assumption of *Archimedean Outcomes*, which says that every outcome in $\mathcal{O}$ has a well-defined utility (in effect ignoring the possibility of infinitely good or bad outcomes and of infinitesimally good or bad outcomes).

DTU is a strong theory, and one in which at least a very large proportion of good decision-theoretic reasoning can be carried out. Here, this will be illustrated by pointing out three consequences of the theory which make its overall structure clear.²⁰

The first consequence is the principle of *Statewise Dominance*:

Statewise dominance If it is certain that p would yield an outcome at least as good as q would, then $p \succeq q$. If there is also a nonzero probability that p yields a strictly better outcome than q , then $p \succ q$.²¹

¹⁹ So-named by Quirk and Saposnik (1962). Stochastic Dominance is equivalent (in the presence of the other principles, including Archimedean Outcomes) to Stochastic Equivalence plus the mostly uncontroversial principle of Statewise Dominance, stated below.

²⁰ By a “consequence of DTU” we mean a sentence that can be derived from DTU plus enough pure mathematics for real analysis. The background pure mathematical theory will be left implicit since the details don’t matter as long as it is sufficiently strong and provably consistent.

²¹ We derive Statewise Dominance from Stochastic Dominance as follows. First, let p and q satisfy the antecedent of the first part of Statewise Dominance, and suppose for contradiction that p is worse than q . By stochastic dominance, it follows that for some o , q is more likely than p is to yield an outcome better than o . Then, conditional on the claim that q would yield an outcome better than o , there must be a nonzero probability that p would yield an outcome no better than o . Hence there is a nonzero probability that there is an outcome better than o that q yields and an outcome no better than o that p yields, so there is a nonzero probability that p yields an outcome worse than what q yields, contradicting the supposition.

Statewise Dominance is a desirable consequence because it is such an overwhelmingly plausible principle, but it has particular significance in the present context, where it has been suggested that the acceptor of Unbounded Utility must give up dominance principles altogether in light of the various paradoxes. Arntzenius, Elga, and Hawthorne (2004) put this idea in a box:

In infinite cases, rationality does not require one to choose one's dominant options

The box can be interpreted as disavowing principles like Countable Sure-Thing, or as disavowing even principles like Statewise or Stochastic Dominance. The consistency of DTU demonstrates, confirming the view of Dietrich and List (2005), that on the second interpretation Arntzenius and coauthors would be overreacting. The consistency also makes clearer how various dominance-like forms of reasoning must be distinguished to avoid paradox.

The second important consequence of DTU is the fact that, since it includes Stochastic Dominance, the comparison of prospects can be simplified by instead comparing the probability distributions over outcomes that they give rise to. In fact, since we are supposing every outcome to have a utility and every utility to be had by some outcome, we can simplify matter further by comparing only the probability distribution over *utilities* (i.e., real numbers) that the two prospects give rise to. That is, the prospect p can be harmlessly conflated with the probability distribution μ that, for each set of real numbers X , assigns to X the probability that p yields an outcome whose utility is in X . Moreover, Prospect Richness guarantees that every real-valued probability distribution corresponds to some prospect, so the question of how to rank prospects reduces entirely to that of how to rank real-valued probability distributions.

The principles of DTU carry over smoothly to the comparison of probability distributions. We use Greek letters μ, ν, ξ as variables for real-valued probability distributions and abuse notation by writing $\mu \succeq \nu$ for the claim that a prospect whose utility is distributed according to μ is at least as good as any prospect that yields utility according to the distribution ν . In this setting the Sure-Thing principle reduces to von Neumann and Morgenstern's (1944) principle of *Independence*:

Independence $\mu \succeq \nu$ if and only if, for any distribution ξ and number $x \in (0, 1)$, $x\xi + (1 - x)\mu \succeq x\xi + (1 - x)\nu$ (the mixture of distributions $x\xi + (1 - x)\mu$ is defined as the distribution satisfying

$$(x\xi + (1 - x)\mu)(X) = x\xi(X) + (1 - x)\mu(X)$$

for any measurable set X).

Stochastic Dominance can be rewritten in terms of distributions as follows:

Stochastic Dominance If for every real number x , μ assigns probability at least as great as ν does to the ray $[x, \infty)$ (i.e., if $\mu[x, \infty) \geq \nu[x, \infty)$), and there is also a real number y such that μ assigns a strictly greater probability to $[y, \infty)$ than ν does ($\mu[y, \infty) > \nu[y, \infty)$), then $\mu \succ \nu$.

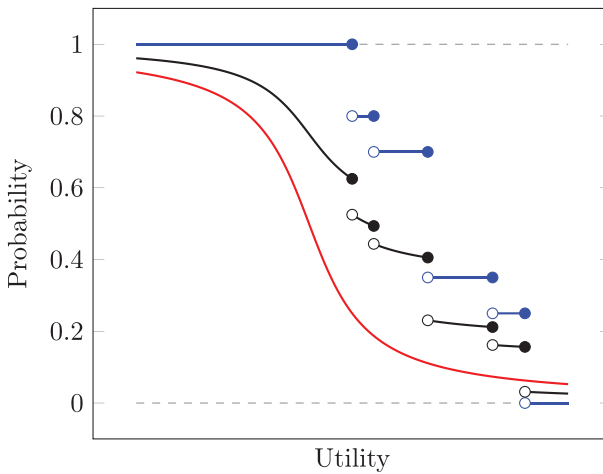


FIGURE 2 Black is the mixture of blue and red (by 0.5). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/nous.12473)]

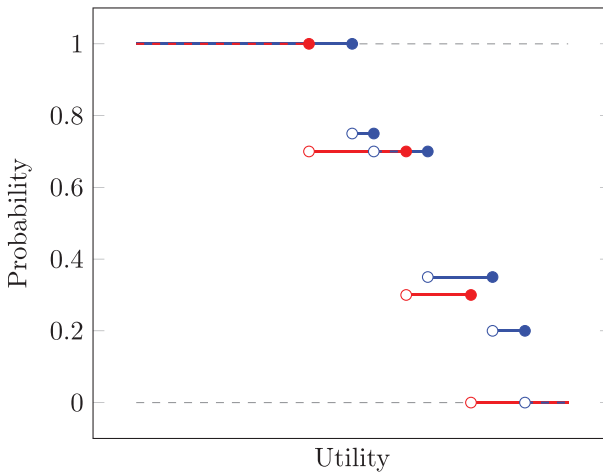


FIGURE 3 Blue dominates red. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/nous.12473)]

Both of these principles can now be visualized in terms of the *cumulative distribution functions*, or CDFs, of real-valued probability distributions. The CDF of μ is the function that assigns to each number x the probability that μ gives to the ray $[x, \infty)$. See Figures 2 and 3 for how the Independence and Stochastic Dominance conditions are visualized in terms of CDFs.

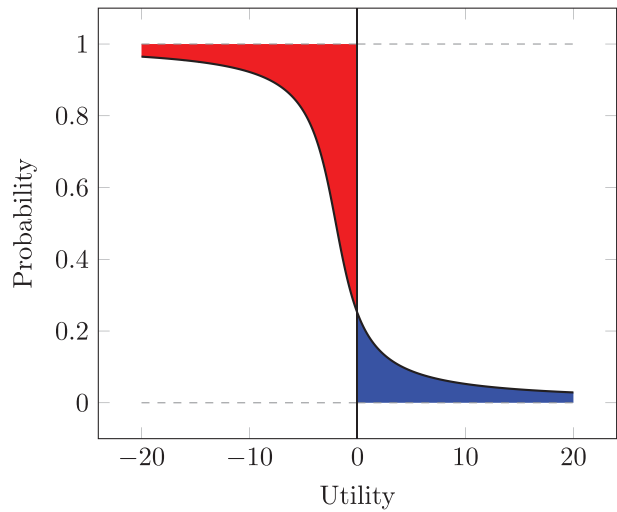
The final important consequence of DTU to discuss is the extent to which it vindicates standard decision-theoretic reasoning in the case of simple prospects (prospects which have only finitely many possible outcomes). By slightly modifying von Neumann and Morgenstern's representation theorem (1944),²² one can show that expected utility theory for simple prospects is simply a theorem of DTU. That is, we can recover the following principle:

Simple EUT If p and q are simple prospects, then $p \succeq q$ if and only if the expected utility of p is at least as great as that of q .

Expected utility can also be visualized in terms of CDFs, as in Figure 4. DTU therefore explains

²² See Joyce (1999) for exposition.

FIGURE 4 The expectation of the distribution is equal to the blue area minus the red area, if both areas are finite. [Color figure can be viewed at wileyonlinelibrary.com]



why expectation-based reasoning is good in the finite cases without falling silent in the infinite cases, for the principles of DTU are fully general. In total, then, DTU is a strong and desirable theory. The question that remains is whether it contains a contradiction.

3.3 | Consistency of DTU

No further contradiction arises in DTU without the addition of Countable Sure-Thing principle:

Theorem 1. *DTU is consistent.*

In this sense we can be said to have solved the motivating paradox. The consistency of DTU is established in Appendix B. In this section the main ideas will be covered by establishing the consistency of a weaker theory, DU, which is DTU but with the axiom of Totality weakened to Restricted Totality (§2.3). DU weakens DTU by withholding on whether every pair of prospects is comparable. The proof of the consistency of DU is intuitive and entirely elementary; and it forms the core of the proof of Theorem 1, which complicates the argument by a standard application of non-principal ultrafilters. The proof strategy in either case can be attributed to Lauwers (Lauwers (2016)), who uses it to establish the consistency of weak fragments of DU and DTU.²³

The existence of non-principal ultrafilters requires the axiom of choice, so the proof of consistency is not acceptable from the perspective of constructive mathematics. This is significant because various authors in economics and philosophy have suggested that non-constructive mathematics should not be used in ethical theorising²⁴ (note that by “constructive” mathematics these authors generally mean a theory like ZF which excludes the axiom of choice but includes the law of excluded middle, which in other contexts is often taken to be a “non-constructive” principle).

²³ In particular, Lauwers can be credited with noticing that the preordering $\sqsubseteq$ defined in this section has properties corresponding to the axioms of Preordering and Independence, and that it can be extended to a total ordering with an ultrafilter as is done in Appendix B.

²⁴ See Fleurbaey and Michel (2003), Zame (2007), Lauwers (2010), Lauwers (2012), Lauwers (2016), Litak (2018), Askell (2018) (p. 257).

Happily, for the consistency of DTU the use of non-constructive mathematics is dispensable by a celebrated result of Gödel (1938).²⁵ However, those who wish to avoid non-constructive mathematics in ethical theorising will find the addition of Totality to DU unacceptable, because Lauwers (2016) shows that DTU already entails non-constructive mathematical principles.

Nevertheless, the consistency of DTU over just DU may be of interest even to those who reject the axiom of Totality, whether they do so to avoid non-constructive mathematics or otherwise. This is because authors who accept some or all of the axioms of DU—including Savage, as well as Joyce (1999), Easwaran (2014a) and many others who reject Totality—take the principles of DU to be not just true principles governing the structure of value under risk, but as principles constraining all coherent ways of preferring prospects. That is, they take (some or all of) the axioms of DU to be true when ‘ $\succeq$ ’ is replaced with a variable universally quantified over all “coherent preference orderings”. On this picture, the consistency of adding the following principle about coherent preference orderings is equivalent to the consistency of DTU:

Coherent Extensibility

- (i) For every pair of prospects p, q and coherent preference ordering $\succeq$ for which $q \not\succeq p$ and $p \not\succeq q$, there is a coherent preference ordering $\succeq'$ that extends $\succeq$ such that either $p \succeq' q$ or $q \succeq' p$.²⁶
- (ii) For any set of coherent preference orderings that extend one another, there is a coherent preference ordering that extends all of them.²⁷

Coherent Extensibility is plausible even supposing Totality fails, for reasons presented in Kaplan (1983) and Hare (2010) (see also de Finetti (1972) §3.10.7 for analogous considerations in probability theory). It is also seemingly acceptable from a constructive perspective: see Remark 1 in Appendix B.

The remainder of this section establishes the following:

Theorem 2. *DU is consistent.*

The consistency will be shown by showing that the theory is true on an interpretation, by constructing an ordering $\sqsubseteq$ of prospects that has the properties DU says $\succeq$ has. What this amounts to is constructing a preordering of probability distributions satisfying Independence, Stochastic Dominance, and Simple EUT (§3.2), which we will also write $\sqsubseteq$.²⁸ See Figures 2, 3, and 4 for diagrams illustrating these conditions.

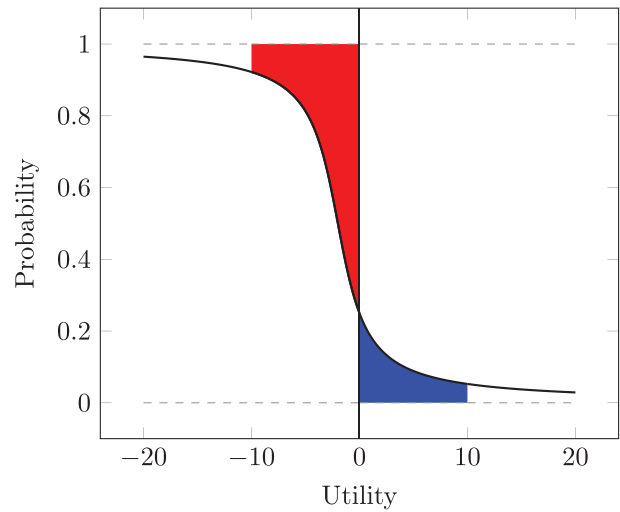
²⁵ For a modern exposition of the result see Jech (2006) pp. 188–190. Gödel showed that ZF—which the previously cited authors who impose constructiveness accept—can prove the consistency of fragments of ZF + Choice that are sufficient for Theorem 1, and so ZF is sufficient for the consistency of DTU.

²⁶ $\succeq_1$ extends $\succeq_2$ if for any p, q , $p \succeq_1 q$ if $p \succeq_2 q$, and $p \not\succeq_1 q$ only if $q \not\succeq_2 p$. Note that every ordering extends itself.

²⁷ Coherent Extensibility implies the existence of a total coherent preference ordering given Zorn’s lemma, so the ‘ $\succeq$ ’ in DTU may be interpreted as that coherent preference ordering.

²⁸ Simple EUT immediately implies Unbounded Utility and Restricted Totality. Independence implies Sure-Thing. Prospect Richness is built into the model because all probability distributions are in the domain of $\sqsubseteq$.

FIGURE 5 The expectation truncated to 10 of the distribution (i.e., $E_t\mu$ for $t = 10$) is equal to the blue area minus the red area. Notice that both areas will be finite for any t . [Color figure can be viewed at wileyonlinelibrary.com]



The construction of $\sqsupseteq$ is built on the observation that the ordering of distributions by their expected values satisfies all three constraints when we restrict attention to distributions whose expectation is well-defined (all of which are confirmed by inspecting the figures).²⁹

The only problem is that the expectation of a distribution is not always defined. This, we have already observed (§1), is the case for the distribution of the St Petersburg prospect. For the construction of $\sqsupseteq$, what we will do instead is compare prospects by the sequence of their *truncated expectations*. For a positive real number t , write $E_t\mu$ for the expectation of the distribution that is just like μ but that puts all the probability of a value bigger than t onto t , and all the probability of a value less than $-t$ onto $-t$. This distribution agrees with μ on subsets of $(-t, t)$, but gives probabilities $\mu([t, \infty))$ and $\mu((-\infty, -t])$ respectively to the utilities t and $-t$.^{30, 31} See Figure 5 for a geometrical representation.

When the expectation of a distribution μ is well-defined, it is given by the limit of the truncated expectations as t goes to infinity. Crucially, truncated expectations are all well-defined for any real-

²⁹ Writing $E\mu$ for the expectation of μ , we observe the following:

- Whenever Simple EUT requires that $\mu \sqsupseteq \nu$, by definition the expectations $E\mu$ and $E\nu$ are well-defined and $E\mu \geq E\nu$.
- The expectation of a mixture of prospects with well-defined expectations is the mixture of their expectations:

$$E(x\xi + (1-x)\mu) = xE\xi + (1-x)E\mu$$

Hence $E(x\xi + (1-x)\mu)$ is at least as great as $E(x\xi + (1-x)\nu)$ if and only if $E\mu$ is at least as great as $E\nu$, when the expectations exist.

- If μ stochastically dominates ν and both have well-defined expectations, $E\mu$ is at least as great as $E\nu$, and $E\mu$ is strictly greater if it strictly stochastically dominates.

³⁰ Easwaran (2014b) appeals to a very subtly different notion of truncation of a prospect. For Easwaran, the truncation of μ to t reassigns all the probability of utility better than t or worse than $-t$ to 0, rather than to t or $-t$. Easwaran (pp. 490–491) shows his notion of truncation to have some pathological properties for prospects that have what he calls “unstable tails”, so it will not work for present purposes.

³¹

$$E_t\mu := \int_{-t}^t z d\mu(z)$$

valued probability distribution. Even if the limit is not well-defined, each truncated expectation is. Thus, *every* pair of probability distributions can be compared by comparing their truncated expectations, and this is how we will build the required preordering.

$\mu \sqsupseteq \nu$ will be defined as: as t increases, eventually $\mathbf{E}_t \mu$ is always at least as great as $\mathbf{E}_t \nu$. More precisely:

$$\mu \sqsupseteq \nu := (\exists s > 0)(\forall t > s)(\mathbf{E}_t \mu \geq \mathbf{E}_t \nu)$$

$\sqsupseteq$ is clearly reflexive, and is transitive because if $\mathbf{E}_t \mu \geq \mathbf{E}_t \nu$ for all t greater than s_1 and $\mathbf{E}_t \mu \geq \mathbf{E}_t \xi$ for all t greater than s_2 , then $\mathbf{E}_t \mu \geq \mathbf{E}_t \xi$ for all t greater than both s_1 and s_2 .

We then use the properties of expectations to show that $\sqsupseteq$ always satisfies Simple EUT, Independence, and Stochastic Dominance:

Simple EUT: If μ is simple, then there is a biggest utility, s_μ , that gets a nonzero probability. Then for t bigger than s_μ , $\mathbf{E}_t \mu = \mathbf{E} \mu$. So in particular, if $\mathbf{E} \mu \geq \mathbf{E} \nu$ then for some s (whichever is larger out of s_μ or s_ν), $\mathbf{E}_t \mu \geq \mathbf{E}_t \nu$ for all $t > s$.

Independence: The truncated expectation of a mixture is the mixture of truncated expectations:

$$\mathbf{E}_t(x\xi + (1-x)\mu) = x\mathbf{E}_t\xi + (1-x)\mathbf{E}_t\mu$$

And by basic algebra we have $\mathbf{E}_t \mu \geq \mathbf{E}_t \nu$ if and only if

$$x\mathbf{E}_t\xi + (1-x)\mathbf{E}_t\mu \geq x\mathbf{E}_t\xi + (1-x)\mathbf{E}_t\nu$$

So $\mu \sqsupseteq \nu$ if and only if $x\xi + (1-x)\mu \sqsupseteq x\xi + (1-x)\nu$.

Stochastic Dominance: If μ stochastically dominates ν , then the truncation of μ to t also stochastically dominates the truncation of ν to t . Hence $\mathbf{E}_t \mu \geq \mathbf{E}_t \nu$ for any t so $\mu \sqsupseteq \nu$. If in addition, for some number s , μ gives a strictly higher probability than ν to (s, ∞) , then $\mathbf{E}_t \mu > \mathbf{E}_t \nu$ for all $t > s$. So $\mu \sqsupset \nu$.

DU is therefore consistent.

4 | FURTHER DIRECTIONS

DTU is a strong and plausible theory, so is a good starting point for the investigation of unbounded utility. Nevertheless, DTU leaves many questions undecided. The model construction techniques for DTU and DU are fairly general, and slight variations in the models show how various comparisons of prospects that might seem obvious are in fact independent of the theory. One diagnosis is that DTU should be extended so as to imply these comparisons. Another is that, in light of the failure of Countable Sure-Thing, we should be a little less trusting of our intuitions when unbounded utility is on the table, so comparisons that go beyond DTU should, sometimes surprisingly, be regarded as suspicious. In this section, we consider two possible extensions of DTU as case studies. These case studies illustrate how there is still a great deal of work, both formal and philosophical, to be done in the investigation of unbounded utility.

4.1 | Expected utility theory

Consider the principle that lies at the center of decision theory:

Expected Utility Theory (EUT) If p and q both have an expected utility, then $p \succeq q$ if and only if the expected utility of p is at least as great as the expected utility of q .

EUT is silent about prospects, like the St Petersburg prospect, which lack an expected utility. Still, the existence of prospects beyond its remit does not refute EUT as stated above; they simply show that expected utility is not the end of the story. And in almost every contemporary treatment of unbounded utility, EUT is assumed to be true. For example, Colyvan (2008), Easwaran (2008), Seidenfeld et al. (2009), Gwiazda (2014), Easwaran (2014b), Bartha (2016), Lauwers and Vallentyne (2016), and Meacham (2019) all presuppose EUT, and take their task to be to *extend* EUT so as to be able to evaluate prospects without an expected utility (Easwaran (2014a) and Smith (2014) are notable exceptions).

There are some dissenters from EUT, for example Buchak (2013) and Smith (2014). But these theorists also depart radically from orthodox decision theory, rejecting the Sure-Thing principle and *ipso facto* the Independence axiom of DTU. However, even if we take the orthodox approach to decision theory with DTU, EUT builds in a novel assumption. That is, Expected Utility Theory is *not* a consequence of DTU, and hence must be supported by some additional argument.

Rather, DTU vindicates only *Simple EUT* (§3.2), which is the restriction of EUT to simple prospects—those with only finitely many possible outcomes. The proof that DTU is compatible with the failure of EUT is in Appendix B (Theorem 3). An easier exercise is to show that the model just constructed for DU is one where EUT fails. For this, consider a prospect that yields utility n with probability $1/2^n$. This prospect has expected utility given by the sum:

$$\frac{1}{2} \times 1 + \frac{1}{4} \times 2 + \cdots + \frac{1}{2^n} \times n + \cdots = 2$$

By contrast, its truncated expected utility is always strictly less than 2, so this prospect is strictly worse than 2 in the model.

While EUT is not implied by DTU, nor is its negation:

Theorem 3. *EUT is independent of DTU.*³²

Whether it should be accepted is therefore an independent philosophical question. I submit that it should not be. The generalization from Simple EUT to EUT proper is almost exactly analogous to the generalization from Sure-Thing to Countable Sure-Thing, and we have seen that this is exactly the sort of generalization that unbounded utility should make us suspicious of.

To draw out the analogy, consider a prospect p that yields utility y with probability x (among other possibilities). The Sure-Thing principle says that p can be compared with any other prospect as follows: first conditional on p yielding y , then conditional on p not yielding y . Thus, the value contributed by the x chance of utility y is independent of whatever other outcomes p might yield, and so in some sense can simply be added to the total. The definition of utility is chosen so that the

³² Proof in Appendix B.

utility contribution of an x chance of utility y is the product xy , so the value of p can be thought of as xy *plus* the value of p that is contributed by the non- y possibilities.

$$xy + \dots$$

Suppose p is simple, so it has finitely many possible utilities, y_1 , through y_n , which it yields with some probabilities x_1 through x_n . By repeating the above reasoning n times, the Sure-Thing principle says that the value of p is given by the sum:

$$x_1y_1 + \dots + x_ny_n$$

Which is exactly the verdict of Simple EUT.

To derive EUT proper from this sort of reasoning, we need to extend it to non-simple prospects, such as a prospect q that yields utilities in the infinite sequence $y_1, y_2, \dots$ with probabilities $x_1, x_2, \dots$. The Sure-Thing principle says that the value of q is given by the value of the contribution of an x_n chance of y_n plus the value of q conditional on not getting utility y_n , for any n . But since there are infinitely many different possible outcomes, it just does not follow that the value of q is determined by adding these contributions together all at once with the sum

$$x_1y_1 + \dots + x_ny_n + \dots$$

which is what EUT demands.

To get this in the same way as we got Simple EUT from Sure-Thing, we would need a principle that allows us to compute the value of a prospect by dividing it into *infinitely* many pieces, and evaluating those pieces all at once. And the most obvious such principle would be *Countable* Sure-Thing. Since we have just been forced to reject Countable Sure-Thing, EUT should also look very dubious, and certainly should not be considered as being on a par with Simple EUT.

4.2 | Symmetries of value

We now turn to another possible extension of DTU. The idea is that the *at least as good* ordering of prospects should be invariant under certain operations on the prospects. For example, if you add a fixed utility all possible outcomes of two prospects, that shouldn't affect which is better of the two. Similarly, if you multiply the utility of all of their possible outcomes by a fixed positive amount. Writing $\alpha \cdot p + \beta$ for a prospect which yields utility $\alpha x + \beta$ whenever p would yield utility x , we formalize this idea with the following two principle:

Shift Invariance For any real number β , $p \succeq q$ if and only if $p + \beta \succeq q + \beta$.

Scale Invariance For any positive real number α , $p \succeq q$ if and only if $\alpha \cdot p \succeq \alpha \cdot q$.

This idea has come up a few times in the literature, such as in Hájek's (Hájek (2014), pp. 490-491) discussion of Gwiazda (2014), and by Easwaran:³³

³³ More recently, Wilkinson (2022) has marshalled a principle closely related to Scale Invariance (called *Scale Independence*) in defence of Unbounded Utility.

[...] utility is only defined up to a shift and a stretch—there is no well-defined 0 and no well-defined unit in which utility is measured. Thus, it ought to be the case that adding a constant to the utility of every outcome of a gamble affects the overall value of the gamble by adding the same constant, and similarly for multiplying the utilities by a positive constant.³⁴ Easwaran (2014b), p. 490.

Another natural idea in the same family is that replacing utility x with utility $-x$ throughout two prospects reverses their comparison.

Reflection Anti-Invariance $p \succsim q$ if and only if $(-1) \cdot q \succsim (-1) \cdot p$.

These principles are plausible, and have already been put to work in various places. But mere plausibility isn't enough. Without some guarantee that they do not lead to a further paradox, they cannot be accepted. The model theory just laid out above and in the appendix allows us to begin the investigation of these principles. There are two extremely salient questions to ask of each principle. First: does it *follow* from DTU? Second: is it *consistent* with DTU, or some other natural theory like DU?

On the first question, the answer is surprisingly negative in all three cases. That is, none of shift or Scale Invariance, or Reflection Anti-Invariance, can be derived in DTU (see Appendix B). On the other hand, the answer is positive in each case on the question of consistency. In fact, it is consistent with DTU that all three principles be true together (see Goodsell (2023), Chapter 4).

These three principles therefore constitute a promising way of strictly extending DTU. The question of whether to accept them once again becomes a philosophical one, and a good example of how decision theory in the presence of unbounded utility remains an interesting topic for further research.

ACKNOWLEDGEMENTS

Thanks to Jeff Russell, Weng Kin San, John Hawthorne, Mark Colyvan, Kenny Easwaran, to an audience at the Pacific APA, and to various anonymous referees at *Noûs* and other journals for helpful comments and discussion on previous drafts.

REFERENCES

- Arntzenius, F., Elga, A., & Hawthorne, J. (2004). Bayesianism, infinite decisions, and binding. *Mind*, 113(450), 251–283.
- Arntzenius, F., & McCarthy, D. (1997). The two envelope paradox and infinite expectations. *Analysis*, 57(1), 42–50.
- Askell, A. (2018). *Pareto Principles in Infinite Ethics*. PhD thesis, New York University.
- Bales, A., Cohen, D., & Handfield, T. (2014). Decision theory for agents with incomplete preferences. *Australasian Journal of Philosophy*, 92(3), 453–70.
- Bartha, P. (2016). Making do without expectations. *Mind*, 125(499), 799–827.
- Beckstead, N., & Thomas, T. (2021). A paradox for tiny probabilities and enormous values. *GPI Working Papers*, (7).
- Blackwell, D. A., & Girshick, M. A. (1979). *Theory of games and statistical decisions*. Courier Corporation.
- Broome, J. (1995). The two-envelope paradox. *Analysis*, 55(1), 6–11.

³⁴ It should be noted that the strengthening of Scale Invariance suggested in a naïve reading of this passage is straightforwardly inconsistent with DTU (using the same argument as in Section 1):

Value Scaling A fifty-fifty gamble between $2 \cdot p$ and 0 is equally good as p .

- Buchak, L. (2013). *Risk and Rationality*. Oxford University Press.
- Chalmers, D. (2002). The st. petersburg two-envelope paradox. *Analysis*, 62(2), 155–157.
- Colyvan, M. (2008). Relative expectation theory. *Journal of Philosophy*, 105(1), 37–44.
- de Finetti, B. (1972). *Probability, Induction, and Statistics*. New York: John Wiley.
- Dietrich, F., & List, C. (2005). The two-envelope paradox: An axiomatic approach. *Mind*, 114(454), 239–248.
- Easwaran, K. (2008). Strong and weak expectations. *Mind*, 117(467), 633–641.
- Easwaran, K. (2014a). Decision theory without representation theorems. *Philosophers' Imprint*, 14.
- Easwaran, K. (2014b). Principal values and weak expectations. *Mind*, 123(490), 517–531.
- Fine, T. (2008). Evaluating the pasadena, altadena, and st petersburg gambles. *Mind*, 117(467), 613–632.
- Fleurbaey, M., & Michel, P. (2003). Intertemporal equity and the extension of the ramsey criterion. *Journal of Mathematical Economics*, 39(7), 777–802.
- Gödel, K. (1938). The consistency of the axiom of choice and of the generalized continuum-hypothesis. *Proceedings of the National Academy of Sciences*, 24(12), 556–557.
- Goodsell, Z. (2023). *Unbounded Utility*. PhD thesis, University of Southern California.
- Gwiazda, J. (2014). Orderly expectations. *Mind*, 123(490), 503–516.
- Hájek, A. (2014). Unexpected expectations. *Mind*, 123(490), 533–567.
- Hare, C. (2010). Take the sugar. *Analysis*, 70(2), 237–247.
- Jech, T. (2006). *Set Theory: The Third Millennium Edition, Revised and Expanded*. Springer Monographs in Mathematics. Springer Berlin Heidelberg.
- Joyce, J. M. (1999). *The Foundations of Causal Decision Theory*. Cambridge University Press.
- Kaplan, M. (1983). Decision theory as philosophy. *Philosophy of Science*, 50(4), 549–577.
- Lauwers, L. (2010). Ordering infinite utility streams comes at the cost of a non-ramsey set. *Journal of Mathematical Economics*, 46(1), 32–37.
- Lauwers, L. (2012). Intergenerational equity, efficiency, and constructibility. *Economic Theory*, 49(2), 227–242.
- Lauwers, L. (2016). Why decision theory remains constructively incomplete. *Mind*, 125(500), 1033–1043.
- Lauwers, L., & Vallentyne, P. (2016). Decision theory without finite standard expected value. *Economics and Philosophy*, 32(3), 383–407.
- Lauwers, L., & Vallentyne, P. (2017). A tree can make a difference. *Journal of Philosophy*, 114(1), 33–42.
- Litak, T. (2018). Infinite populations, choice and determinacy. *Studia Logica: An International Journal for Symbolic Logic*, 106(5), 969–999.
- McGee, V. (1999). An airtight dutch book. *Analysis*, 59(4), 257–265.
- Meacham, C. J. G. (2019). Difference minimizing theory. *Ergo: An Open Access Journal of Philosophy*, 6.
- Nalebuff, B. (1989). Puzzles: The other person's envelope is always greener. *The Journal of Economic Perspectives*, 3(1), 171–181.
- Nover, H., & Hájek, A. (2004). Vexing expectations. *Mind*, 113(450), 237–249.
- Quirk, J. P., & Saposnik, R. (1962). Admissibility and measurable utility functions. *The Review of Economic Studies*, 29(2), 140–146.
- Russell, J. S. (2020). Non-archimedean preferences over countable lotteries. *Journal of Mathematical Economics*, 88(May 2020), 180–186.
- Russell, J. S. (2021). On two arguments for fanaticism. *GPI Working Papers*, (17).
- Russell, J. S., & Isaacs, Y. (2021). Infinite prospects. *Philosophy and Phenomenological Research*, 103(1), 178–198.
- Savage, L. (1954). *The Foundations of Statistics*. Wiley Publications in Statistics.
- Schoenfield, M. (2014). Decision making in the face of parity. *Philosophical Perspectives*, 28(1), 263–277.
- Seidenfeld, T., Schervish, M., & Kadane, J. (2009). Preference for equivalent random variables: A price for unbounded utilities. *Journal of Mathematical Economics*, 45, 329–340.
- Smith, N. J. J. (2014). Is evaluative compositionality a requirement of rationality? *Mind*, 123(490), 457–502.
- von Neumann, J., & Morgenstern, O. (1944). *Theory of Games and Economic Behavior*. Princeton University Press.
- Wilkinson, H. (2022). In defense of fanaticism. *Ethics*, 132(2), 445–477.
- Zame, W. (2007). Can intergenerational equity be operationalized? *Theoretical Economics*, 2, 187–202.

How to cite this article: Goodsell, Z. (2024). Decision theory unbound. *Noûs*, 58, 669–695. <https://doi.org/10.1111/nous.12473>

APPENDIX A: AXIOMS OF DTU

Preordering *At least as good as* is reflexive ($p \succeq p$) and transitive (if $p \succeq q \succeq r$, then $p \succeq r$).

Totality For any prospects p and q , either $p \succeq q$ or $q \succeq p$.

Sure-Thing Let E be a subset of Ω with probability strictly between zero and one such that p and q are equally good conditional on E (i.e., $p \sim_E q$). Then $p \succeq q$ if and only if $p \succeq_{\Omega \setminus E} q$.

Prospect Richness For any function³⁵ f from Ω to $\mathcal{O}$, there is a prospect in $\mathcal{P}$ that yields $f(s)$ when the coin tosses land in the sequence s .

Unbounded Utility For any real number x , there is an outcome in $\mathcal{O}$ that has utility x (§2.1).

Archimedean Outcomes Every outcome in $\mathcal{O}$ has a utility.

Stochastic Dominance If for every outcome o , p is at least as likely as q is to yield an outcome better than o , then p is at least as good as q . If there is also some outcome o such that p is strictly more likely than q to yield an outcome better than o , then p is strictly better than q .

DU replaces Totality with:

Restricted Totality If p and q are simple, and every pair of outcomes either might yield is comparable, then either $p \succeq q$ or $q \succeq p$.

Also of note are the following consequence of DU.

Independence $\mu \succeq \nu$ if and only if, for any distribution ξ and number $x \in [0, 1]$, $x\xi + (1 - x)\mu \succeq x\xi + (1 - x)\nu$.

Simple EUT If p and q are simple prospects, then $p \succeq q$ if and only if the expected utility of p is at least as great as that of q .

APPENDIX B: PROOF OF CONSISTENCY OF DTU

We now prove the main result (Theorem 1) and some other theorems. We proceed by extending the proof of Section 3.3 with an ultrafilter. It is also possible to give a more abstract proof by adapting the technique of Fine (2008) to accommodate Stochastic Dominance and applying Zorn's lemma directly, however the concrete models generated by the truncated expectation ordering are particularly natural (they are, for example, anticipated by Easwaran (2014b) and Lauwers (2016)) and

³⁵ Strictly, we should restrict to *measurable* functions, where Ω is given its usual probability space structure and $\mathcal{O}$ is given measurable space structure compatible with the ordering induced by $\succeq$ on constant functions from Ω to $\mathcal{O}$.

straightforward, and are sufficiently modular to achieve all desired independence and consistency results.

Definition 1 (Cobounded ultrafilter). A *cobounded ultrafilter* on $[0, \infty)$ is a set of subsets of $[0, \infty)$, $\mathcal{U}$, such that:

- If $X \subseteq Y$ and $X \in \mathcal{U}$, then $Y \in \mathcal{U}$.
- If $X \in \mathcal{U}$ and $Y \in \mathcal{U}$, then $X \cap Y \in \mathcal{U}$.
- For every $X \subseteq [0, \infty)$, either $X \in \mathcal{U}$ or $[0, \infty) \setminus X \in \mathcal{U}$.
- $\mathcal{U}$ contains no bounded sets.

Lemma 4 (Ultrafilter lemma). *There is a cobounded ultrafilter.*

Proof. Standard (e.g., Jech (2006), Theorem 7.5). □

In what follows, let $\mathcal{U}$ be a cobounded ultrafilter that is held fixed throughout. Roughly speaking, we will extend $\sqsupseteq$ by adding $p \sqsupseteq q$ when the set of $t \in \mathbf{R}^+$ for which the t -truncated expected utility of p exceeds the t -truncated expected utility of q for all t is a member of $\mathcal{U}$. The following standard definition will make it easier to show that the resultant ordering has the desired properties:

Definition 2. For functions f and g from $[0, \infty)$ to $\mathbf{R}$, say $f \sim_{\mathcal{U}}^* g$ if f and g are the same on a member of $\mathcal{U}$. Let the set of *hyperreals* be the set of equivalence classes of functions from $[0, \infty)$ to $\mathbf{R}$ modulo $\sim_{\mathcal{U}}^*$. For such a function f , let $[f]$ be the corresponding hyperreal number.

Lemma 5. *The hyperreals form an ordered field under pointwise addition, pointwise multiplication, and the total ordering $\geq$, defined so that $[f] \geq [g]$ if and only if the set of points x for which $f(x) \geq g(x)$ is in $\mathcal{U}$.*

Proof. Standard (see also Lemma 6). □

Theorem 1. *DTU is consistent.*

Proof. We will modify the model defined for Theorem 2 by replacing $\sqsupseteq$ with $\sqsupseteq^*$, defined as:

$$\mu \sqsupseteq^* \nu := [\mathbf{E}_t(\mu)] \geq [\mathbf{E}_t(\nu)]$$

Now we check that each of the desired principles holds in the model. Simple EUT holds for the same reasoning as before. It remains to check that $\sqsupseteq^*$ is a total preordering that satisfies Independence and Stochastic Dominance.

Totality and Preordering. Lemma 5.

Independence. For the same reasons as in the proof of Theorem 2, we have:

$$\mathbf{E}_t(x\xi + (1-x)\mu) = x\mathbf{E}_t\xi + (1-x)\mathbf{E}_t\mu$$

And by Lemma 5 we have

$$[x\mathbf{E}_t\xi + (1-x)\mathbf{E}_t\mu] = x[\mathbf{E}_t\xi] + (1-x)[\mathbf{E}_t\mu]$$

Now, $\mu \sqsupset^* \nu$ is equivalent to $[\mathbf{E}_t \mu] \geq [\mathbf{E}_t \nu]$, and so is also equivalent to

$$[\mathbf{E}_t(x\xi + (1-x)\mu)] \geq [\mathbf{E}_t(x\xi + (1-x)\nu)],$$

which is equivalent to $x\xi + (1-x)\mu \sqsupset^* x\xi + (1-x)\nu$.

Stochastic Dominance. For Theorem 2 we showed that if μ dominates ν , then $\mathbf{E}_t \mu \geq \mathbf{E}_t \nu$ for all t in $[0, \infty)$. And $[0, \infty) \in \mathcal{U}$, so $\mu \sqsupset^* \nu$. And if μ strictly dominates then for some $s > 0$, $\mathbf{E}_t \mu > \mathbf{E}_t \nu$ for all $t > s$, and the set of points greater than s is in $\mathcal{U}$ because it is cobounded, so $\mu \sqsupset^* \nu$. $\square$

Remark 1. A natural model of the theory of DU with Coherent Extensibility (§3.3) can be constructed in ZF using a similar technique. This establishes that the theory is not non-constructive in the sense of Lauwers (2016). For an arbitrary cobounded filter F , let $\sqsupset_F$ be $\sqsupset^*$ as in the proof of Theorem 1 but with F instead of $\mathcal{U}$. Then we may interpret ‘coherent preference ordering’ as being a preordering of the form $\sqsupset_F$ for some cobounded filter F . That each $\sqsupset_F$ satisfies DU is the same argument as the proof of Theorem 1, and Coherent Extensibility is true in the resulting model because cobounded filters are extendible and the union of cobounded filters that mutually extend each other is also a cobounded filter.

Theorem 3. *EUT is independent of DTU.*³⁶

This theorem requires some more definitions.

Definition 3 (Continuous Hyperreal). For functions $f : [0, \infty) \rightarrow \mathbf{R}$ and $g : [0, \infty) \rightarrow \mathbf{R}$, say that $f \sim_{\mathcal{U}}^c g$ if, for any $\varepsilon > 0$, the set of points x for which $|f(x) - g(x)| < \varepsilon$ is a member of $\mathcal{U}$. Then, let the *continuous hyperreals* be the set of functions modulo $\sim_{\mathcal{U}}^c$. Let $[f]^c$ be the continuous hyperreal of which f is a member. The reals can be identified with the equivalence classes of constant functions $[y \mapsto x]^c$.

Lemma 6. *The continuous hyperreals form an ordered ring under pointwise addition, pointwise multiplication, and the total ordering $\leq$, defined so that $[f]^c \leq [g]^c$ if and only if for every positive real number ε , the set of points x for which $f(x)$ is no greater than $g(x) + \varepsilon$ is a member of $\mathcal{U}$.*

Proof. First we must check that the pointwise operations are well-defined on the continuous hyperreals: that if $[f] = [g]$ and $[h] = [i]$, then $[f] + [h] = [g] + [i]$ and $[f] \cdot [h] = [g] \cdot [i]$. To see that this is so, fix some ε and let X and Y respectively be the sets on which f and h and g and i are within $\varepsilon/2$ of each other in the case of addition, and within $\sqrt{\varepsilon}$ in the case of multiplication. Then, since $X \cap Y$ is a member of $\mathcal{U}$, the operations work the same on different members of the same equivalence class. That the continuous hyperreals form a ring follows from the fact that the pointwise operations make the underlying set of functions into a ring.

That $\leq$ is reflexive is obvious, and that $\leq$ is transitive follows from the fact that if $X \in \mathcal{U}$ is the set on which $f + \varepsilon/2$ is at least as great as g , and $Y \in \mathcal{U}$ is the set on which $g + \varepsilon/2$ is as great as h , then $f + \varepsilon$ is at least as great as h on $X \cap Y$, which is a member of $\mathcal{U}$. Totality follows from the maximality property of ultrafilters. The ordered ring properties follow from the properties of real addition and multiplication. $\square$

³⁶ Proof in Appendix B.

Proof of Theorem 3.. To show that EUT does not follow from DTU, notice that the ordering in Theorem 1 extends the ordering in Theorem 2, so also violates EUT.

To show that EUT is consistent, we use the same proof as Theorem 1 but with $\sqsubseteq^*$ replaced with $\sqsubseteq^c$, defined as:

$$\mu \sqsubseteq^c \nu := [\mathbf{E}_t \mu]^c \geq [\mathbf{E}_t \nu]^c$$

We then prove that the resulting model satisfies the desired principles. Things go the same as before with one exception, which is Stochastic Dominance. For this, it is slightly harder to show that when μ strictly dominates ν it is strictly better, because in general $\mathbf{E}_t \mu$ can always strictly exceed $\mathbf{E}_t \nu$, but by a vanishing amount, in which case $[\mathbf{E}_t \mu]^c = [\mathbf{E}_t \nu]^c$. However in the case where μ strictly dominates ν , there is some utility s such that μ is strictly more likely than ν to yield an outcome at least as good as s . It follows that for $t > s$, there is some $\delta > 0$ such that $\mathbf{E}_t \mu > \mathbf{E}_t \nu + \delta$, so in fact $[\mathbf{E}_t \mu]^c > [\mathbf{E}_t \nu]^c$ as required.

It remains only to show that EUT holds in the model. This corresponds to the condition that when μ and ν have expectations, $\mu \sqsubseteq^c \nu$ if and only if $\mathbf{E} \mu \geq \mathbf{E} \nu$. So suppose that μ does have an expectation. Then

$$\mathbf{E} \mu := \lim_{x \rightarrow \infty} \mathbf{E}_x \mu$$

Which is to say that, for all $\varepsilon > 0$, there is an $s > 0$ such that for $t > s$, $|\mathbf{E}_t \mu - \mathbf{E} \mu| < \varepsilon$. Therefore, $[\mathbf{E}_t \mu]^c = \mathbf{E} \mu$, so in general if ν also has an expected utility, then $\mu \sqsubseteq^c \nu$ if and only if $\mathbf{E} \mu \geq \mathbf{E} \nu$. $\square$

Remark 2. For any unbounded subset $X \subseteq [0, \infty)$, there is a cobounded ultrafilter that contains X .

Proof. Jech (2006), Theorem 7.5. $\square$

Corollary 1. *Thus, if $\mathbf{E}_t \mu \geq \mathbf{E}_t \nu$ on an unbounded set, it is consistent with DTU that μ is at least as good as ν .*

Corollary 2. *Since $\mathbf{E}_t \mu \geq \mathbf{E}_t \nu$ on an unbounded set if and only if $\neg(\mu \sqsubseteq \nu)$, it follows that if μ is not at least as good as ν in the model constructed for Theorem 2, it is consistent in DTU that ν is at least as good as μ .*

Remark 3. In the above proofs, $\mathbf{E}_t \mu$ can be replaced with $\mathbf{E}_t^{fg} \mu$, defined as

$$\int_{-g(t)}^{f(t)} z d\mu(z)$$

for any two nonnegative, increasing, and unbounded functions f and g , and the resulting ordering of prospects still satisfies the relevant combination of principles.

Corollary 3. *Let the symmetric St Petersburg prospect be that which yields utilities 2 and -2 each with probability $1/4$, 4 and -4 with probabilities $1/8$, and in general utilities 2^n and -2^n each with*

TABLE 4 The symmetric St Petersburg prospect.

1/4	1/4	1/8	1/8	...	$2^{-(n+1)}$	$2^{-(n+1)}$	...
2	-2	4	-4	...	2^n	-2^n	...

probabilities $1/2^{(n+1)}$ (see Table 4). By setting f and g appropriately, it is possible to build models for Theorem 3 where:

- The symmetric St Petersburg prospect is better than every outcome.
- The symmetric St Petersburg prospect is worse than every outcome.
- For any particular outcome x , the symmetric St Petersburg prospect is equally good as x . (If x isn't 0, then Reflection Anti-Invariance is false in the model.)

Theorem 7. Reflection Anti-Invariance holds in the models constructed for DTU, DU, and DTU + EUT.

Theorem 8. Shift Invariance holds in the models constructed for DU and DTU + EUT.

Proof. For Shift Invariance in the model of DTU + EUT, write $\mu_{+\beta}$ for the distribution that gives $X + \beta$ (i.e., $\{x + \beta : x \in X\}$) the same probability as $\mu(X)$. We have

$$\begin{aligned}
 E_t \mu_{+\beta} &= \int_{-t}^t z d\mu_{+\beta}(z) = \int_{-t-\beta}^{t-\beta} z + \beta d\mu(z) \\
 &= \int_{-t}^t z + \beta d\mu(z) + \int_{-t-\beta}^{-t} z + \beta d\mu(z) - \int_{t-\beta}^t z + \beta d\mu(z) \\
 &= E_t \mu + \int_{-t}^t \beta d\mu(z) + \left(\int_{-t-\beta}^{-t} z d\mu(z) - \int_{t-\beta}^t z d\mu(z) \right)
 \end{aligned}$$

The second term in the last line tends to β , since μ is a probability distribution, and third term tends to 0, because it is bounded in absolute value above by $t\beta\mu[t - \beta, t]$. Hence $[E_t \mu_{+\beta}]^c = [E_t \mu]^c + \beta$. $\square$

Theorem 9. Scale Invariance is true in the model constructed for DU.

Theorem 10. Scale Invariance fails in all models of DTU constructed above.

Proof. Write μ_2 for the distribution satisfying:

$$\mu_2(X) = \mu(X \div 2)$$

For all measurable X . μ_2 is the distribution corresponding to multiplying each possible utility by 2.

Notice that $\mathbf{E}_t \mu_2 = \mathbf{E}_{2t} \mu$. Hence for a counterexample to Scale Invariance in the model it will suffice to find μ and ν such that for some δ , the sets

$$\{t \in \mathbf{R}^+ : \mathbf{E}_t \mu \geq \mathbf{E}_t \nu + \delta\}$$

and

$$\{t \in \mathbf{R}^+ : \mathbf{E}_{2t} \nu \geq \mathbf{E}_{2t} \mu + \delta\}$$

are both in $\mathcal{U}$.

To this end, notice also that a function $f(t) : \mathbf{R}^+ \rightarrow \mathbf{R}^+$ is $\mathbf{E}_t \mu$ for some μ when f is nondecreasing, continuous, differentiable almost everywhere, with a nonincreasing derivative bounded above by 1, which tends towards 0 in the limit. So it suffices to find f and g with this property so that the set on which $f(t) \geq g(t) + \delta$ and the set on which $g(2t) \geq f(2t) + \delta$ are both in $\mathcal{U}$.

Set $f(t)$ and $g(t)$ as follows:

$$f(t) := \sqrt{t+1}$$

$$g(t) := \begin{cases} \sqrt{t+1} & 0 \leq t < 1 \\ \sqrt{t+1} + \frac{\sin(\pi \log_2 t)}{1000} & t \geq 1 \end{cases}$$

We have for $t \geq 1$,

$$g(t) - f(t) = \frac{\sin(\pi \log_2 t)}{1000}$$

$$g(2t) - f(2t) = \frac{\sin(\pi \log_2 2t)}{1000}$$

$$= -\frac{\sin(\pi \log_2 t)}{1000}$$

So for any $\delta > 0$, the set of points $t \geq 1$ for which $f(t) \geq g(t) + \delta$ and $g(2t) \geq f(2t) + \delta$ are the same, and are both unbounded.

Next we must show that $f(t)$ and $g(t)$ are indeed $\mathbf{E}_t \mu$ and $\mathbf{E}_t \nu$ for some distributions μ and ν . For this, notice that both are continuous, and both are smooth everywhere besides 1, with derivatives and second derivatives as follows:

$$f'(t) = \frac{1}{2\sqrt{t+1}}$$

$$f''(t) = \frac{-1}{4\sqrt[3]{t+1}}$$

$$g'(t) = \frac{1}{2\sqrt{t+1}} + \frac{\pi \cos(\pi \log_2 t)}{1000(\ln 2)t}$$

$$g''(t) = \frac{-1}{4\sqrt[3]{t+1}} + \frac{-\pi^2(\ln 2) \sin(\pi \log_2 t) - \pi \cos(\pi \log_2 t)}{1000(\ln 2)t^2}$$

f' is bounded above by 1, tends towards 0, and is positive. f'' is negative so f' is always decreasing, so $f(t)$ is indeed $\mathbf{E}_t\mu$ for some μ . For g , notice that for $t \geq 1$ the numerator of the added term decays like $1/t$ for g' and like $1/t^2$ for g'' , which is much more rapidly than $1/\sqrt{t}$ or $1/\sqrt[3]{t}$, so $g(t)$ is $\mathbf{E}_t\nu$ for some ν .

What we have shown is that for any $\delta > 0$, $f(t) \geq g(t) + \delta$ if and only if $g(2t) \geq f(2t) + \delta$. Hence if the set of such t is in $\mathcal{U}$, we have a counterexample to Scale Invariance. Suppose otherwise, then for every ε the set of t such that $|f(t) - g(t)| < \varepsilon$ is in $\mathcal{U}$. For a given ε , this set is given by

$$\{t \in \mathbf{R}^+ : |\sin(\pi \log_2 t)| < 1000\varepsilon\}$$

Then, we may repeat the above reasoning, now comparing $f(t)$ with $h(t)$, defined as follows:

$$h(t) := \begin{cases} \sqrt{t+1} & 0 \leq t < 2 \\ \sqrt{t+1} + \frac{\cos(\pi \log_2 t)}{1000} & t \geq 2 \end{cases}$$

And by the same argument we must conclude the following set to be in $\mathcal{U}$ if there is no counterexample to Scale Invariance:

$$\{t \in \mathbf{R}^+ : |\cos(\pi \log_2 t)| < 1000\varepsilon\}$$

But $\sin$ and $\cos$ are out of phase: $\sin$ takes a value near 0 when $\cos$ takes an extremal value, and vice-versa. So the two sets are disjoint (for small values of ε). So in any case there is a counterexample to Scale Independence in the model. $\square$